

Binary Search - Traditional + Conditional Based

Traditional

While $L < R$ if $M == T$ return M

Round 3
at index 4

Sorted in
general
ascending
Order

$[-5, -3, -2, 1, 3, 5]$
L M L M R

$$M = (3+5) // 2 = 4$$

$L=0$

$R=N-1$

$$M = (L+R) // 2$$

elif $T > M$

$$L = M+1$$

$$= (0+5) // 2$$

$$\Rightarrow M = 2.5 = 2$$

Lookup 3 = T

if lookup 1 = T

$[-5, -3, -2, 1, 3, 5]$
L M L M R

elif $T \leq M$

$$R = M-1$$

found 1 at index 3

if $L < R$ cross each other then ~~that~~ means that the element does not exist in that array.

alternatively, M can also be

$$M = L + \left[\frac{R-L}{2} \right]$$

used to
avoid integer
overflow

Conditional-Based (over-under)

we want to
find the first
false

F F F T T T
 $[T, T, T, F, F, F]$
0 1 2 3 4 5
L M L M R
2 3 4 5

if $N=1000$

$$\Rightarrow N=500$$

$$\Rightarrow N=250$$

$$\Rightarrow N=125$$

$$\Rightarrow N=62$$

Time: $O(\log_2 N)$

Space: $O(1)$